Few-Shot 3D Brain Tissue Segmentation under Memory Constraints using a U-Net-Based Approach

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1. Introduction

Medical image segmentation plays a fundamental role in modern clinical analysis and computer assisted diagnosis.Deep learning approaches have significantly improved the performance of medical image segmentation systems in recent years. The introduction of U-Net by[4] demonstrated that encoder decoder architectures with skip connections can achieve highly accurate biomedical image segmentation even when limited annotated data are available .This idea was further extended to volumetric segmentation by replacing 2D operations with their 3D counterparts by [2] enabling the network to better capture spatial context in volumetric medical images . More recently, nnU-Net [3]showed that careful optimisation of preprocessing, training strategies, data augmentation and inference procedures can improve segmentation performance without relying on highly complex architectural modifications .

Despite these advances, 3D medical image segmentation remains challenging under limited data and memory constrained conditions. Volumetric MRI data has a computational bottleneck compared to conventional 2D imaging. Due to the cubic scaling of 3D convolutional activations,the full volume training is quite difficult on consumer grade hardware with limited GPU memory (e.g. 16GB).A patch based training and inference strategy was adopted from [2] to capture local context while remaining within the hardware's VRAM constraints.Medical datasets are small and highly imbalanced because certain anatomical structures occupy only a tiny fraction of the image volume .This imbalance can lead to unstable optimisation and poor segmentation performance for minority classes.

In this work, we investigate few-shot 3D brain tissue segmentation on the MRBrainS13 dataset using a modified 3D U-Net framework under data and memory constraints. A baseline 3D U-Net model was first implemented using patch based volumetric training. Several improvements inspired by the nnU-Net framework and recent medical segmentation literature were explored like Instance Normalisation, LeakyReLU activation, Gaussian weighted sliding window inference, foreground aware and class aware patch sampling and data augmentation [3] . Extensive experiments and ablation studies were conducted to analyse how these strategies influence segmentation performance under severe class imbalance conditions.

The experimental results show that inference strategies and sampling methods can improve segmentation stability and overall Dice performance. However, minority tissue classes with extremely small voxel distributions remain difficult to segment.

2. DATASET AND PREPROCESSING

2.1. MRBrainS13 Dataset

The experiments in this work were conducted using the MRBrainS13 dataset which is a publicly available benchmark dataset for brain tissue segmentation in magnetic resonance imaging (MRI). The dataset contains volumetric brain MRI scans together with manually annotated segmentation masks for multiple tissue classes.Figure 1 shows representative slices from the MRI volumes together with their corresponding ground truth segmentation masks.

2.2. Data Exploration and Analysis

Before model training data analysis was performed to understand the characteristics, distribution of classess

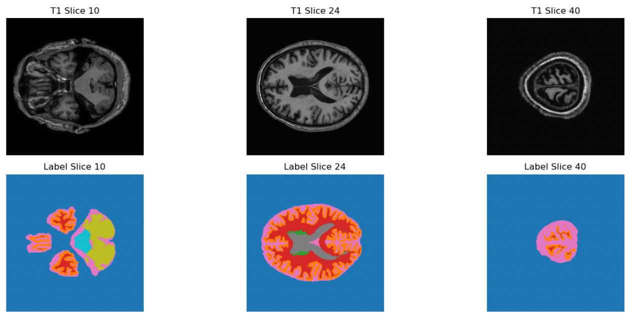

Figure 1. Different axial slices are shown to illustrate anatomical variability across the 3D MRI volume and the complexity of voxel wise brain tissue segmentation.

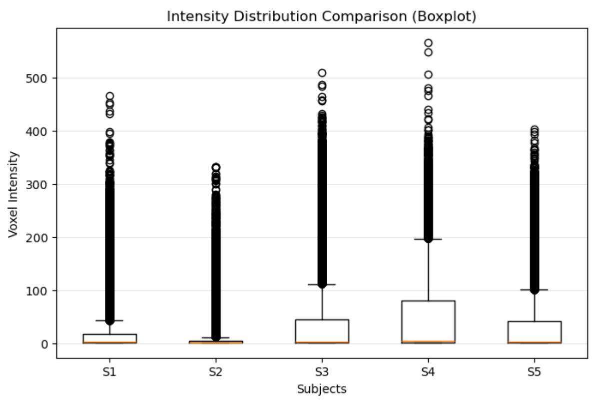

Figure 2. MRI intensity variation across different subjects.

and intensity variation of the MRI volumes and segmentation labels.

MRI intensity values varied significantly across subjects due to scanner acquisition differences and anatomical variability. Figure 2 shows the intensity distributions observed across different subjects.Such variability can negatively affect optimisation during training if left unnormalised.

To address this issue,z-score normalisation was applied independently to each subject using only non-background voxels.This avoids bias introduced by the large number of zero valued background voxels and ensures that normalization reflects true tissue intensity distributions. Let x denote the MRI intensity values and μ and σ represent the mean and standard deviation computed from brain voxels only. The normalised volume was computed using z-score normalisation as:

$$x_{\text{norm}} = \frac{x - \mu}{\sigma + 10^{-8}} \tag{1}$$

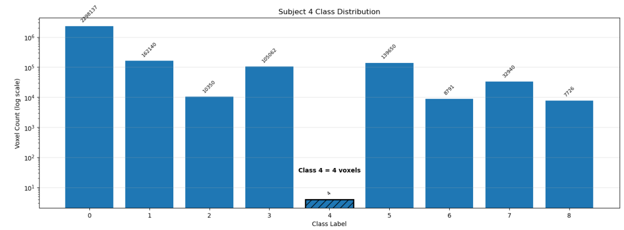

Figure 3. Voxel distribution across segmentation classes in subject 4.

2.3. Class Imbalance Analysis

A major challenge identified during preprocessing was severe class imbalance within the segmentation masks. The background class occupied the majority of voxels while several tissue classes contained comparatively few samples. In particular, Class 4 was found to be extremely sparse across multiple subjects.The most extreme case was found in subject 4 which contains only 4 voxels of class 4. This imbalance was later observed to strongly influence segmentation performances for minority tissue classes.

Figure 3 illustrates the voxel distribution across subject 4 segmentation classes.

3. Methodology

3.1. Baseline 3D U-Net

The baseline model was implemented as a 3D U-Net style architecture, following the general encoder decoder design introduced in U-Net and later extended to volumetric segmentation in 3D U-Net [2, 4]. The network receives a 3D MRI patch as input and predicts a voxel wise segmentation mask with nine output classes.

The encoder consists of repeated 3D convolutional blocks followed by downsampling using max pooling. This contracting path progressively reduces the spatial resolution while increasing the number of feature channels which allows the model to learn increasingly abstract anatomical representations. The feature sizes used in the encoder were [32,64,128].We can also increase these feature size to get better feature maps but it depends upon the computational resources. After the encoder, a bottleneck block was used with twice the number of channels of the final encoder layer. This bottleneck represents the deepest part of the network where the spatial resolution is lowest but the learned feature representation is richest.

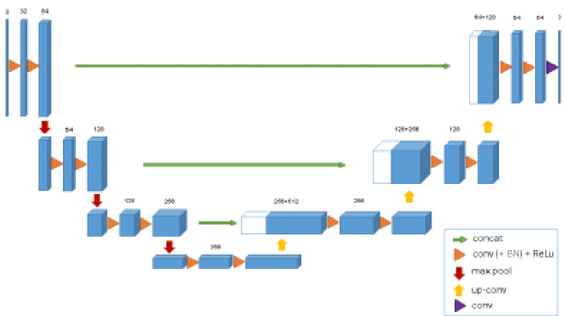

Figure 4. 3D U-Net architecture used as the baseline segmentation model.Figure adapted from [2]

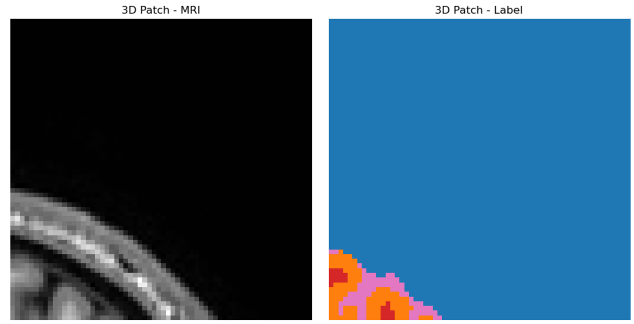

Figure 5. MRI slice extracted through random patch from the MRBrainS13 dataset.

The decoder reconstructs the segmentation map by progressively upsampling the bottleneck features using transposed 3D convolutions. Skip connections concatenate high-resolution encoder features with decoder features at matching scales. These skip connections help preserve spatial detail that may be lost during downsampling and improve localisation accuracy. A final 1×1×1 convolution maps the decoder output to nine class logits.

3.2. Training Setup

In this work, five training subjects from the dataset were used. Each subject contains a 3D T1 weighted MRI volume together with a corresponding voxel wise segmentation label map. The segmentation masks consist of nine classes including background and multiple brain tissue structures. Since the dataset size is extremely limited a Leave-One-Out Cross Validation (LOOCV) protocol was adopted. In each fold, four subjects were used for training while the remaining subject was reserved for validation and testing. This process was repeated five times so that every subject was evaluated exactly once.

The MRI volumes exhibit large spatial dimensions and high memory requirements which makes full volume 3D training impractical on limited GPU hardware that's why patch based training was adopted from [2] throughout this work.This strategy was a solution for the computational bottleneck in the 3d-Unet architecture.

A single MRI volume in the MRBrainS13 dataset ($\approx 240 \times 240 \times 48$) contains approximately 2.76×10^6 voxels. Processing a full volume at float32 precision requires roughly 10.55 MB per feature map. For a 3D U-Net with a base width of 32 filters, the activations for the initial layer alone would occupy 337.5 MB.A standard architecture consists of 15–20 layers.The full volume training would require more than 10 GB's for forward-pass activations which leaves insufficient overhead for backpropagation, gradients and optimizer states.

Patch extraction also increased the effective number of training samples available from the limited dataset. However, random patch sampling introduced additional challenges because many extracted patches contains mostly background voxels. This issue has further amplified the class imbalance problem and motivated the exploration of foreground-aware and class-aware patch sampling methods in later experiments.Figure 5 shows a sample of random patch with their corresponding ground truth segmentation masks.

The baseline experiments initially used Dice loss alone.Dice loss directly optimises the overlap between the predicted segmentation and ground truth masks which makes it effective for small anatomical structures that occupy only a limited number of voxels.

After establishing the baseline, a combined Dice and Cross Entropy loss was used which was inspired by [3] to improve voxel wise class discrimination while preserving overlap based optimisation. The combined loss function was defined as:

$$\mathcal{L}_{DiceCE} = \mathcal{L}_{Dice} + 0.3\mathcal{L}_{CE} \tag{2}$$

To study the effect of Cross Entropy weighting ,an ablation study was conducted (see Section 4).

The model was trained using 3D patches of size $96 \times 96 \times 32$ with 100 patches. Patch density and patch size were later explored through ablation experiments. A batch size of 1 was used due to the high memory requirements of volumetric 3D convolutions. Each experiment was trained for 20 epochs using the AdamW optimiser with a learning rate of 10^{-3} and weight decay of 10^{-4}. AdamW was selected because it provides stable optimisation for deep neural networks while weight decay helps reduce overfitting in the few-shot training setting. Segmentation performance was evaluated using the Dice Similarity Coefficient which measures the overlap between predicted segmentation voxels and ground truth annotations. Dice score is widely used in medical image segmentation because it is robust to class imbalance and directly reflects segmentation quality.

$$Dice = \frac{2|P \cap G|}{|P| + |G|} \tag{3}$$

where P represents the set of predicted voxels and G represents the ground truth voxels.

3.3. Best Model Architecture

Our optimized model utilizes a 3D U-Net backbone with **Instance Norm** and **LeakyReLU** activations ($slope = 0.01$) and **Gaussian weighted sliding window inference** . Based on our ablation studies (see Section 4), the final configuration employs a patch size of $96^2 \times 32$ and combined dice and cross entropy loss.

4. Experiments and Ablation Studies

4.1. Architecture and Inference Improvements

The baseline model used the 3D U-Net architecture with random patch sampling of $64 \times 64 \times 32$ with 50 patches and standard Dice loss. Later on, improvements were made by keeping these parameters same patch size= $96 \times 96 \times 32$ with 100 patches and combined dice and cross entropy loss to study the effect of each improvement. The strongest improvement was obtained by replacing BatchNorm3d and ReLU with InstanceNorm3d and LeakyReLU followed by Gaussian-weighted sliding-window inference. This change was motivated from the nnU-Net [3].The author found that for 3D U-Nets with small batches, you should normalize every image based on its own current statistics even during the testing phase. InstanceNorm computes statistics per sample and is therefore more suitable for low batch 3D medical segmentation.Figure 6 and 7 shows that instance norm is more stable than batch norm

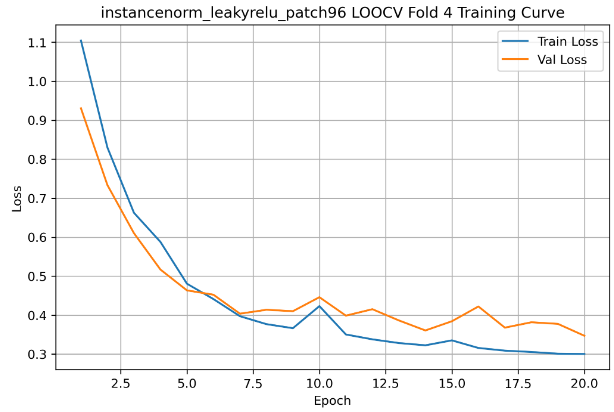

Figure 6. Instance norm with leaky Relu loss curve.

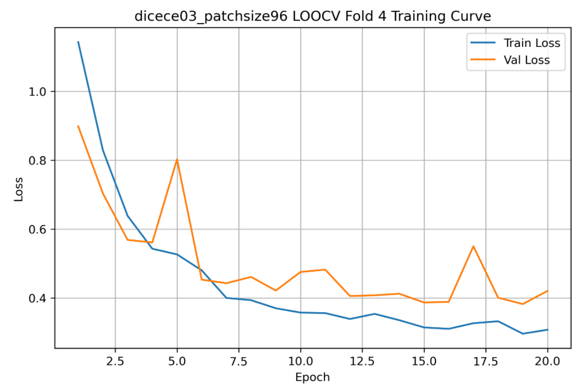

Figure 7. Batch norm with relu loss curves.

Table 1. Architecture and inference improvements.

Method	Mean Dice	Class 4
Patch96 DiceCE	0.803	0.259
InstanceNorm + LeakyReLU	0.820	0.329
+ Gaussian inference	0.825	0.354

As we are not using residual connections to migitate zero or vanishing gradients so leaky relu is a good alternate. Standard patch based inference often suffers from stitching artifacts or edge bias.The model predictions are less accurate at the borders of a patch due to a lack of surrounding context. This motivated our adoption of Gaussian weighted sliding-window inference which assigns higher confidence to the central voxels of a patch during the reconstruction of the final volume which leads to smoother transitions and improved anatomical fidelity.

These results suggest that optimisation stability and

Table 2. Class imbalance strategies compared with the best model.

Method	Mean Dice	Class 4
Baseline model	0.557	-
Best model	0.825	0.354
Foreground-aware patching	0.823	0.327
Class-4-aware patching	0.825	0.341
Data augmentation	0.811	0.340
Focal loss	0.808	0.329
Final combined system	0.815	0.363

inference aggregation are important for 3D patch-based segmentation. However, Class 4 remained difficult especially for Subject 4 where the Dice score remained zero due to the extreme scarcity of Class 4 voxels.

4.2. Class Imbalance Experiments

Several imbalance aware strategies were evaluated, including weighted CE loss, foreground-aware patching, class-aware patching, data augmentation, and Focal Loss. Table 2 summarises these results.

The final combined system achieved the highest Class 4 Dice score of 0.363 but reduced the overall mean Dice to 0.815. This shows a trade off between rare class sensitivity and overall segmentation accuracy. Foreground aware and class aware sampling improved exposure to foreground structures but did not solve the most extreme rare class failure. The augmentation pipeline included random flipping, intensity scaling, intensity shifting, Gaussian noise, and elastic deformation. These augmentations were motivated by both the original U-Net framework [4] and recent medical imaging augmentation reviews [1] which highlight augmentation as an important strategy for reducing overfitting in small medical datasets.However, augmentation did not improve the final overall segmentation performance in this work. Although some minority-class cases showed slight improvement.Similarly, Focal Loss reduced overall performance and did not improve Class 4 compared with the best model.

These results suggest that the remaining failure is not caused only by design or sampling choices. Instead, it is strongly related to the extreme scarcity of annotated voxels for Class 4. Subject 4 contains only four voxels belonging to Class 4 which makes reliable segmentation statistically unstable even when imbalance aware methods are applied.

Table 3. Effect of Cross Entropy weighting in the DiceCE loss.

Class	Dice only	CE=0.2	CE=0.3
0	0.988	0.991	0.992
1	0.778	0.807	0.813
2	0.251	0.627	0.496
3	0.801	0.846	0.843
4	0.109	0.167	0.233
5	0.690	0.717	0.743
6	0.815	0.820	0.871
7	0.729	0.798	0.809
8	0.159	0.657	0.555

4.3. Dice and Cross Entropy Loss Ablation

After establishing the Dice only baseline, a combined Dice and Cross Entropy loss was evaluated.Two CE weights were tested: 0.2 and 0.3.

Table 3 compares the Dice only baseline with the two DiceCE configurations. Both CE weighted variants improved substantially over Dice loss alone. The CE=0.2 setting improved several smaller classes particularly Class 2 and Class 8. The CE=0.3 setting produced more balanced performance across most classes and improved Class 4 from 0.109 to 0.233.

These results show that the balance between Dice and Cross Entropy is task dependent. A lower CE weight preserved stronger sensitivity to some smaller structures while CE=0.3 provided more consistent performance across the segmentation classes. Therefore, DiceCE with CE=0.3 was selected for subsequent experiments.

4.4. Patch Sampling Density Ablation

Patch sampling density was evaluated by varying the number of random training patches sampled per epoch. Table 4 shows that increasing the number of patches per epoch consistently improved mean Dice. The mean Dice increased from 0.557 with 50 patches to 0.732 with 150 patches.

The improvement suggests that too few patches provide insufficient coverage of the 3D volume under few-shot conditions. Increasing the number of patches exposes the network to more anatomical variation and reduces underfitting. However, this comes at the cost of

Table 4. Patch sampling density ablation.

Patches/epoch	Mean Dice	Time (min)	GPU MB
50	0.557	2.19	329.99
100	0.642	3.74	345.21
150	0.732	5.21	333.01

Table 5. Patch size ablation.

Patch size	Mean Dice	Time (min)	GPU MB
$64 \times 64 \times 32$	0.642	3.74	345.21
$80 \times 80 \times 32$	0.776	3.51	489.11
$96 \times 96 \times 32$	0.803	3.46	658.49

increased training time. GPU memory remained almost unchanged because the patch size and batch size were fixed, confirming that patch density affects optimisation time rather than instantaneous memory usage.

4.5. Patch Size Ablation

Patch size was also found to strongly affect segmentation performance. Larger patches provide more anatomical context which is important for distinguishing structures in volumetric MRI. Table 5 shows that increasing patch size from $64 \times 64 \times 32$ to $96 \times 96 \times 32$ improved mean Dice from 0.642 to 0.803.

The result confirms that spatial context is important in 3D segmentation. However, the increase in memory usage shows the trade-off between context and computational cost. Despite the memory increase, the $96 \times 96 \times 32$ patch size remained feasible on the available GPU and was therefore used for subsequent experiments.

5. Discussions

5.1. The Normalization Bottleneck

A critical finding was the failure of **Batch Normalization** when using a batch size of 1. Because the model calculated statistics from only a single 3D patch which causes the "heartbeat" spikes observed in the training and validation curves (see Figure 7). By switching to **Instance Normalization** we normalized each image independently. This removed the dependency on batch size and allowed the model to converge smoothly despite the memory constraints.

5.2. The Sparsity Problem of Class 4

A critical diagnostic finding was the model's performance on Class 4 which represents the most extreme case of class imbalance in the MRBrainS13 dataset.Subject 4 consists of only four voxels. Under these conditions, the Dice coefficient is hypersensitive to spatial jitter. The structure is so small that the model only needs to miss by a single voxel to get a score of zero. This shows that the Dice metric is simply too sensitive to measure success for such small anatomical structures.Subject 4 highlights a limitation of the evaluation metric.

6. Conclusion

This study addressed few-shot 3D brain tissue segmentation under 16GB VRAM constraints. We demonstrated that a baseline 3D U-Net can be improved (+26.8% Mean Dice) through a data centric combination of **Instance Normalization**, **Gaussian-weighted inference**, and **hybrid DiceCE loss**.

Our analysis concludes that while architectural stability and patch-based sampling successfully addressed hardware bottlenecks but the extreme class imbalance of sparse structures remains a data level limitation. Specifically, the 0.0 Dice score on Subject 4 (4 voxels) underscores a limitation of overlap based metrics. Dice is size dependent and fails to reward near misses in such singular structures. Future work should evaluate localization accuracy using **Hausdorff Distance** [5] and explore **Attention Gates** [6] to further refine feature calibration. Ultimately, our model achieves a Mean Dice of **0.825**, providing a robust framework for volumetric analysis in resource-constrained clinical environments.

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